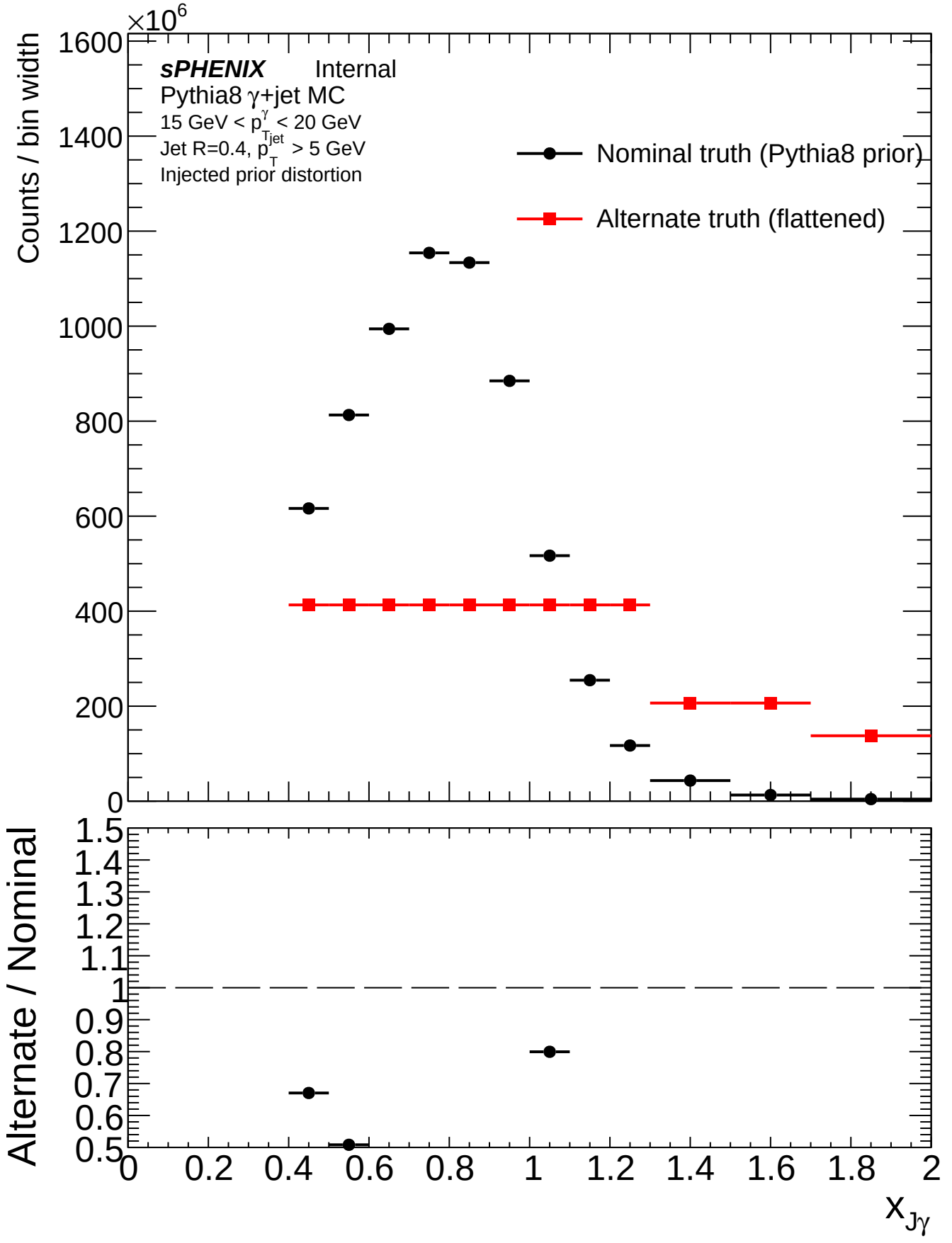
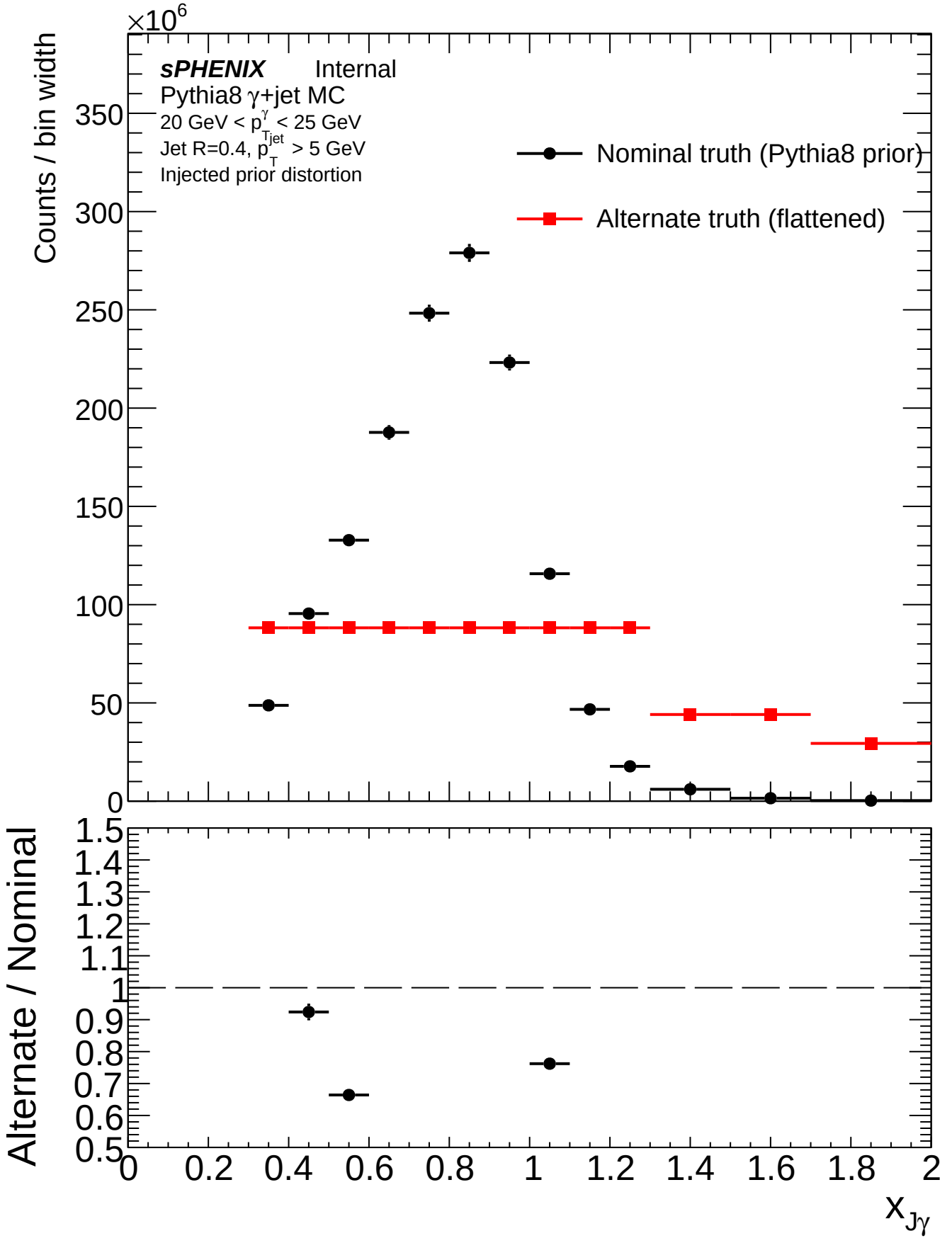
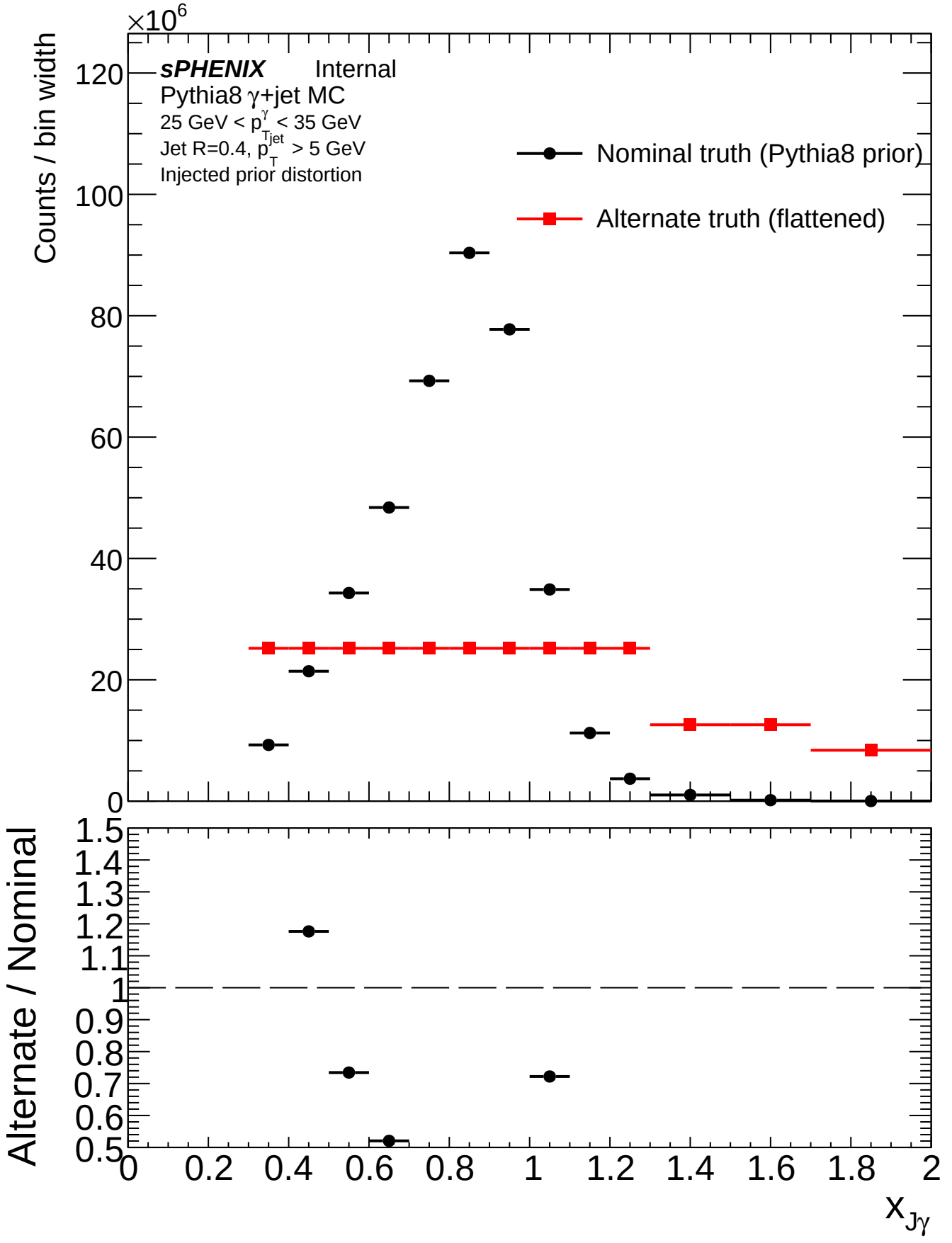
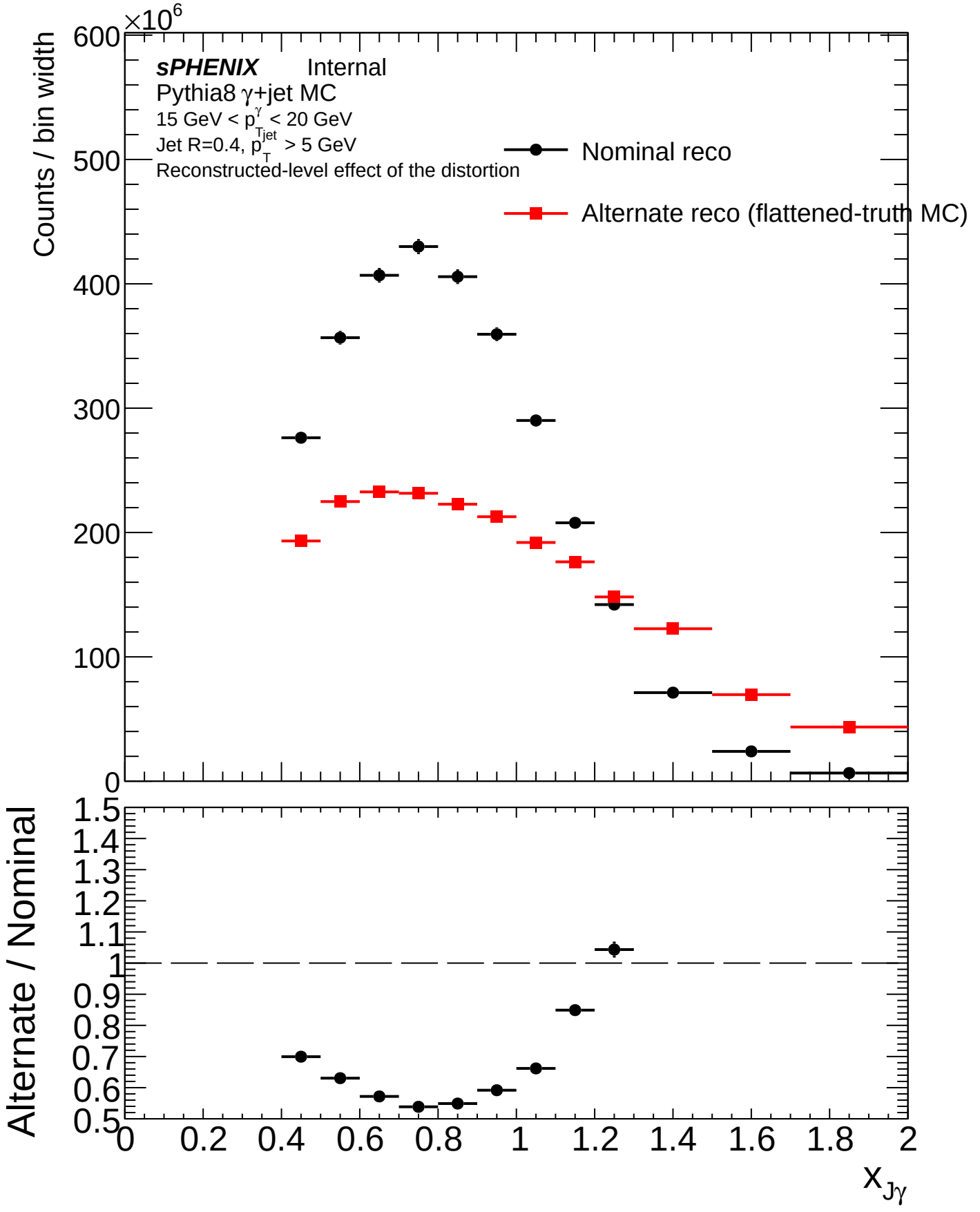
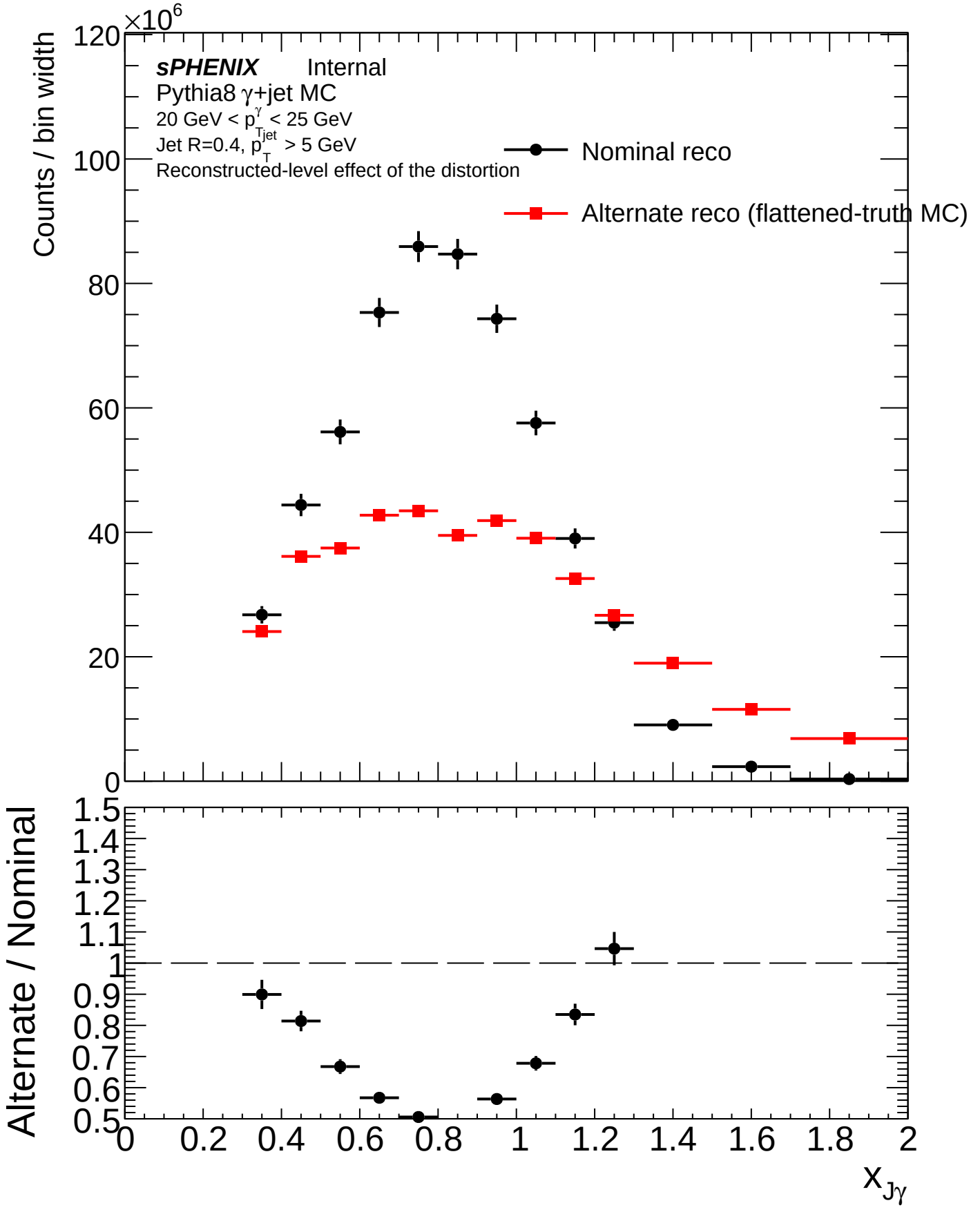
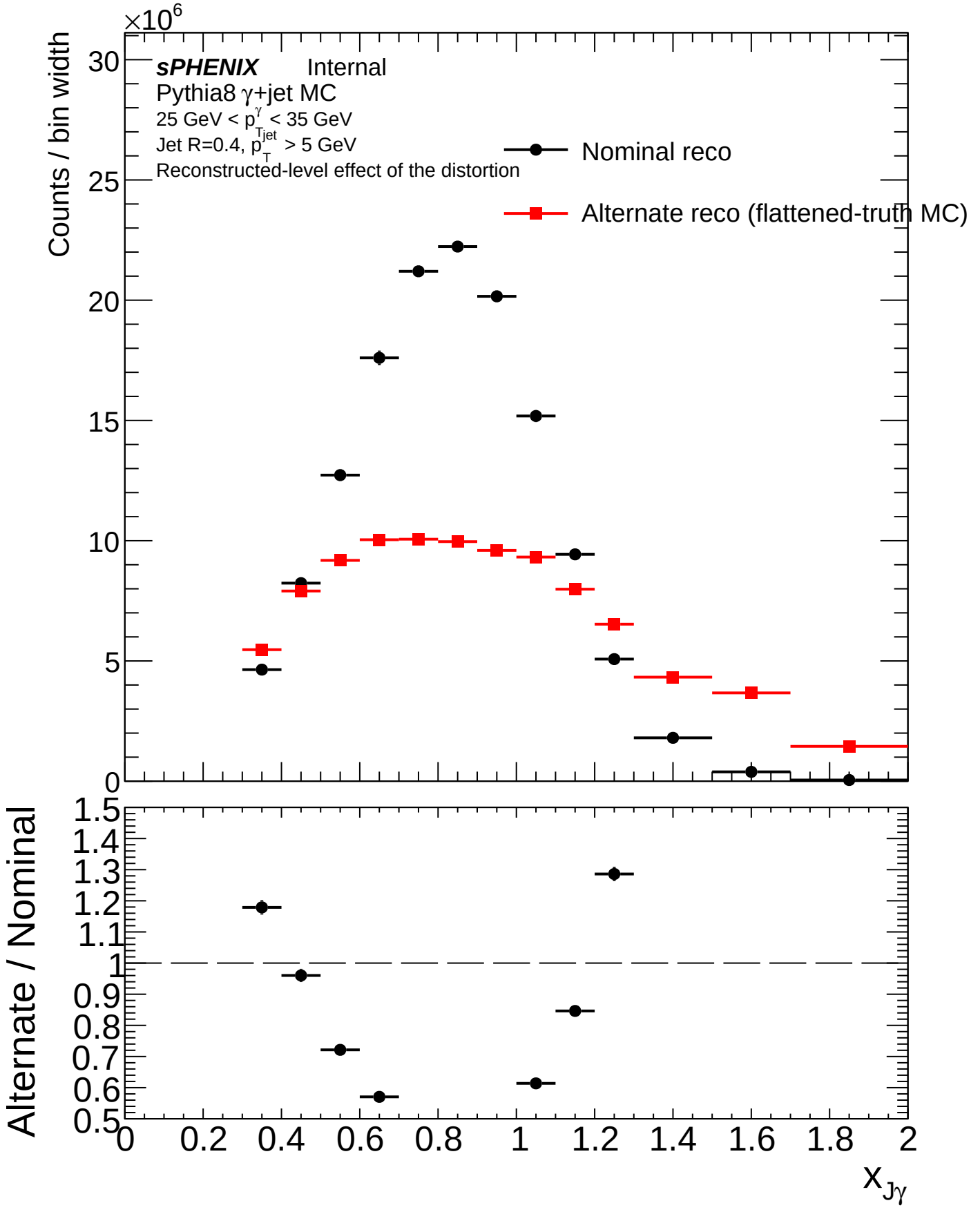

×10^6
sPHENIX Internal
Pythia8 γ+jet MC
15 GeV < p_T^γ < 20 GeV
Jet R=0.4, p_T^Tjet > 5 GeV
Injected prior distortion
Nominal truth (Pythia8 prior)
Alternate truth (flattened)
Counts / bin width
Alternate / Nominal
x_Jγ

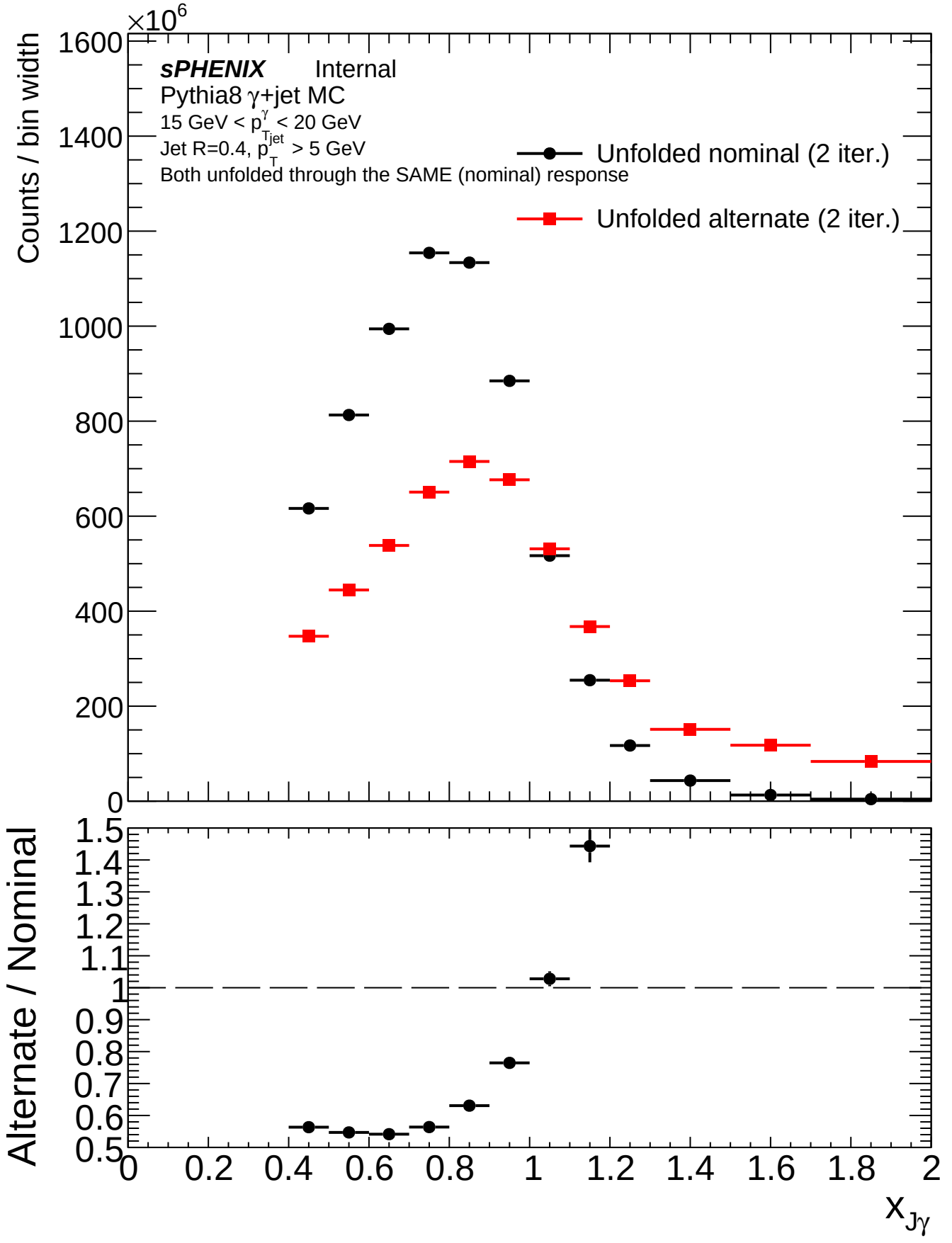

×10^6
sPHENIX Internal
Pythia8 γ+jet MC
20 GeV < p_T^γ < 25 GeV
Jet R=0.4, p_T^Tjet > 5 GeV
Injected prior distortion
Nominal truth (Pythia8 prior)
Alternate truth (flattened)
Counts / bin width
Alternate / Nominal
x_Jγ

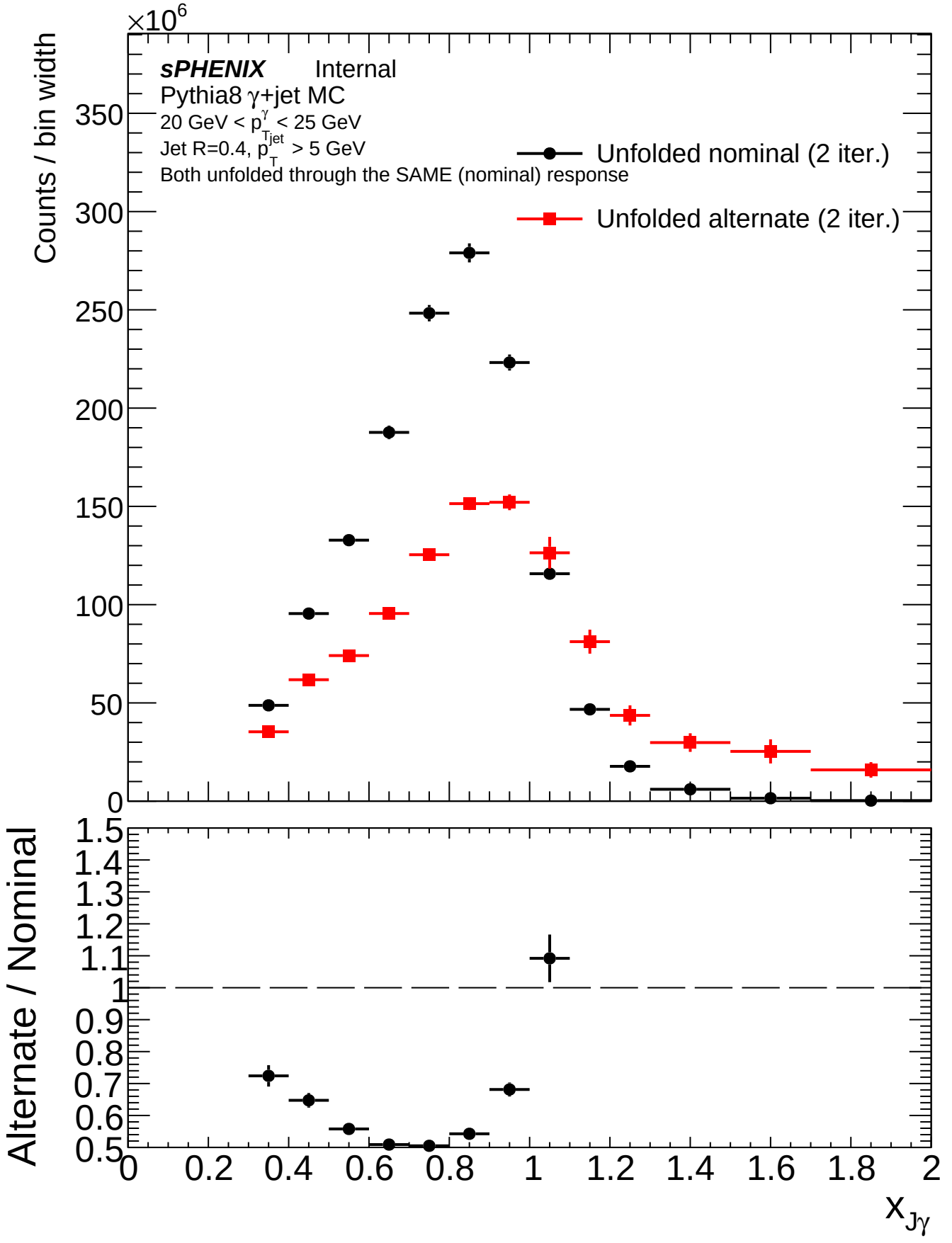

×10^6
sPHENIX Internal
Pythia8 γ+jet MC
25 GeV < p_T^γ < 35 GeV
Jet R=0.4, p_T^Tjet > 5 GeV
Injected prior distortion
Nominal truth (Pythia8 prior)
Alternate truth (flattened)
Counts / bin width
Alternate / Nominal
x_Jγ

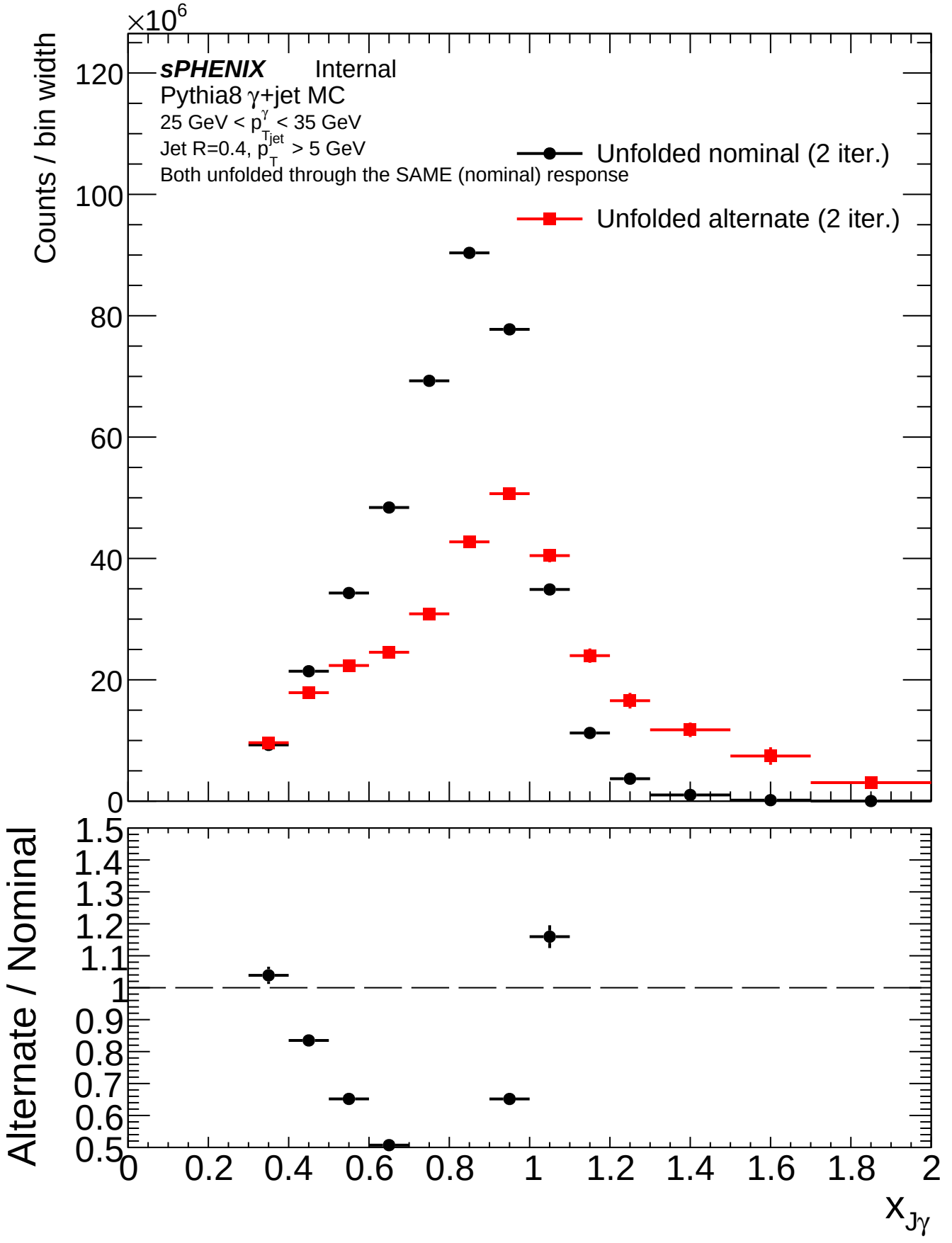

×10^6
600
500
400
300
200
100
0
Counts / bin width
sPHENIX Internal
Pythia8 γ+jet MC
15 GeV < p_T^γ < 20 GeV
Jet R=0.4, p_T^Tjet > 5 GeV
Reconstructed-level effect of the distortion
Nominal reco
Alternate reco (flattened-truth MC)
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
0.6
0.5
Alternate / Nominal
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
x_Jγ

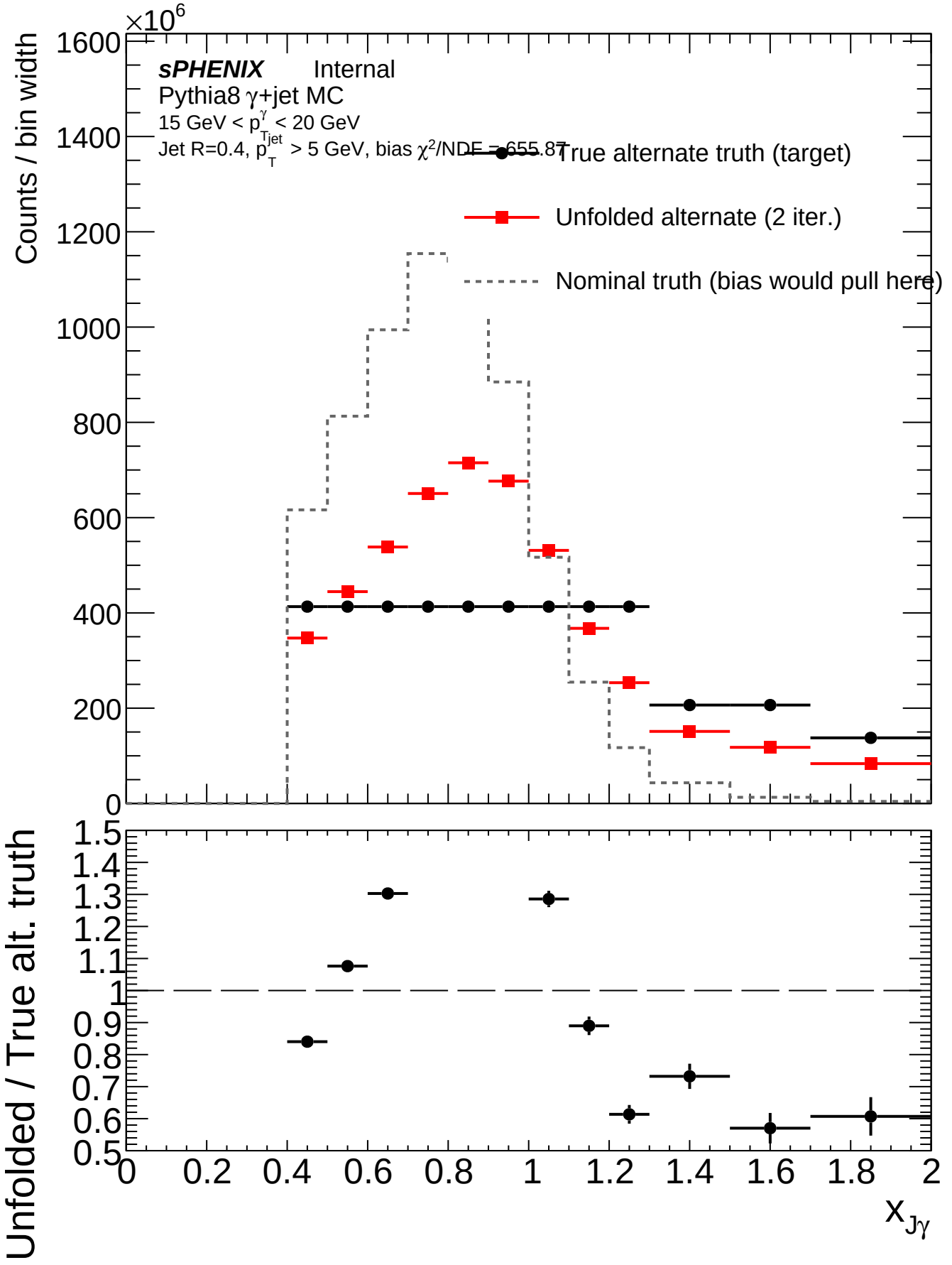

×10^6
120
100
80
60
40
20
0
Counts / bin width
sPHENIX Internal
Pythia8 γ+jet MC
20 GeV < p_T^γ < 25 GeV
Jet R=0.4, p_T^Tjet > 5 GeV
Reconstructed-level effect of the distortion
Nominal reco
Alternate reco (flattened-truth MC)
1.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7
0.6
0.5
Alternate / Nominal
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
x_Jγ

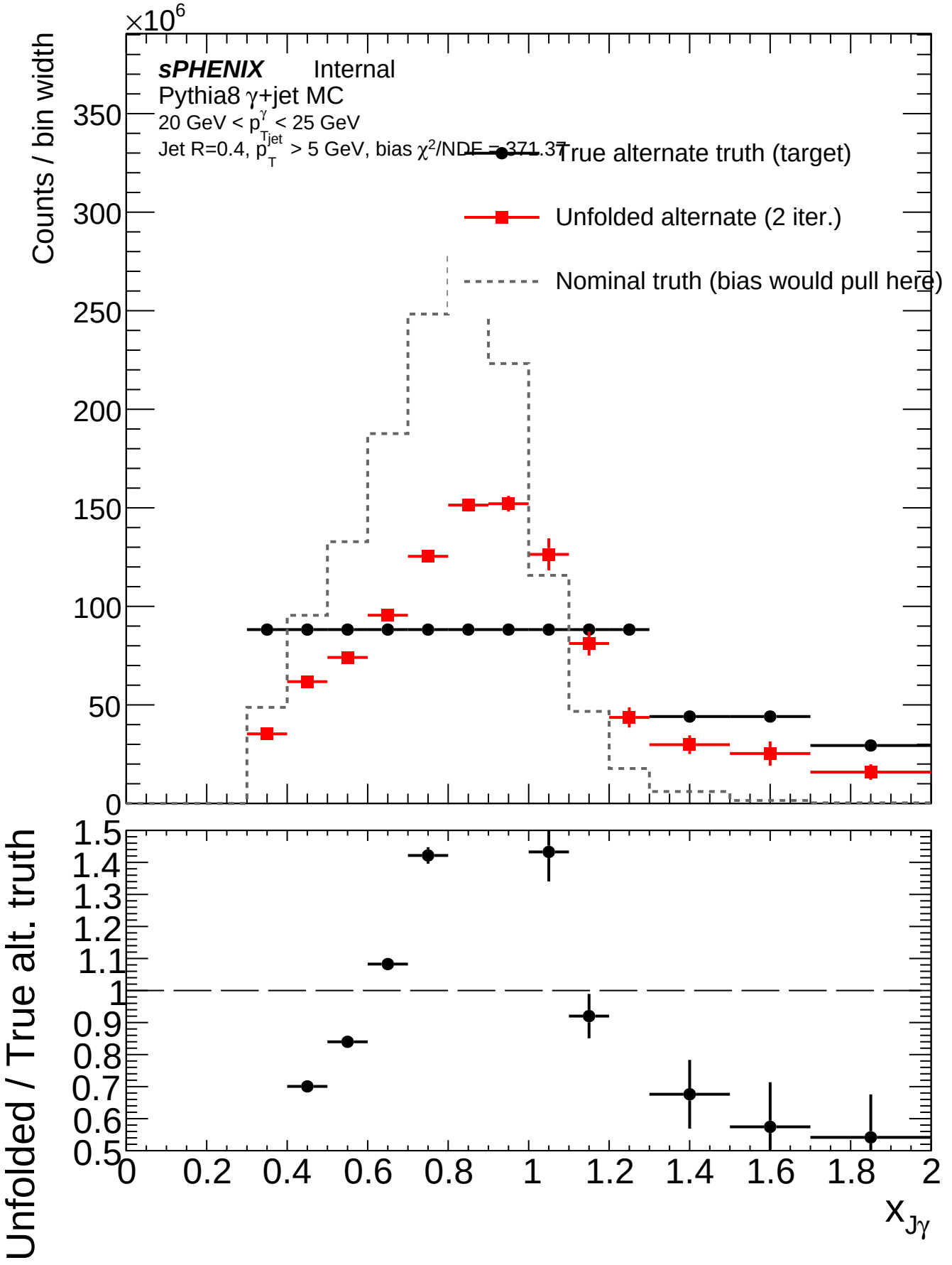

×10^6
Counts / bin width
sPHENIX Internal
Pythia8 γ+jet MC
25 GeV < p_T^γ < 35 GeV
Jet R=0.4, p_T^Tjet > 5 GeV
Reconstructed-level effect of the distortion
Nominal reco
Alternate reco (flattened-truth MC)
Alternate / Nominal
x_Jγ

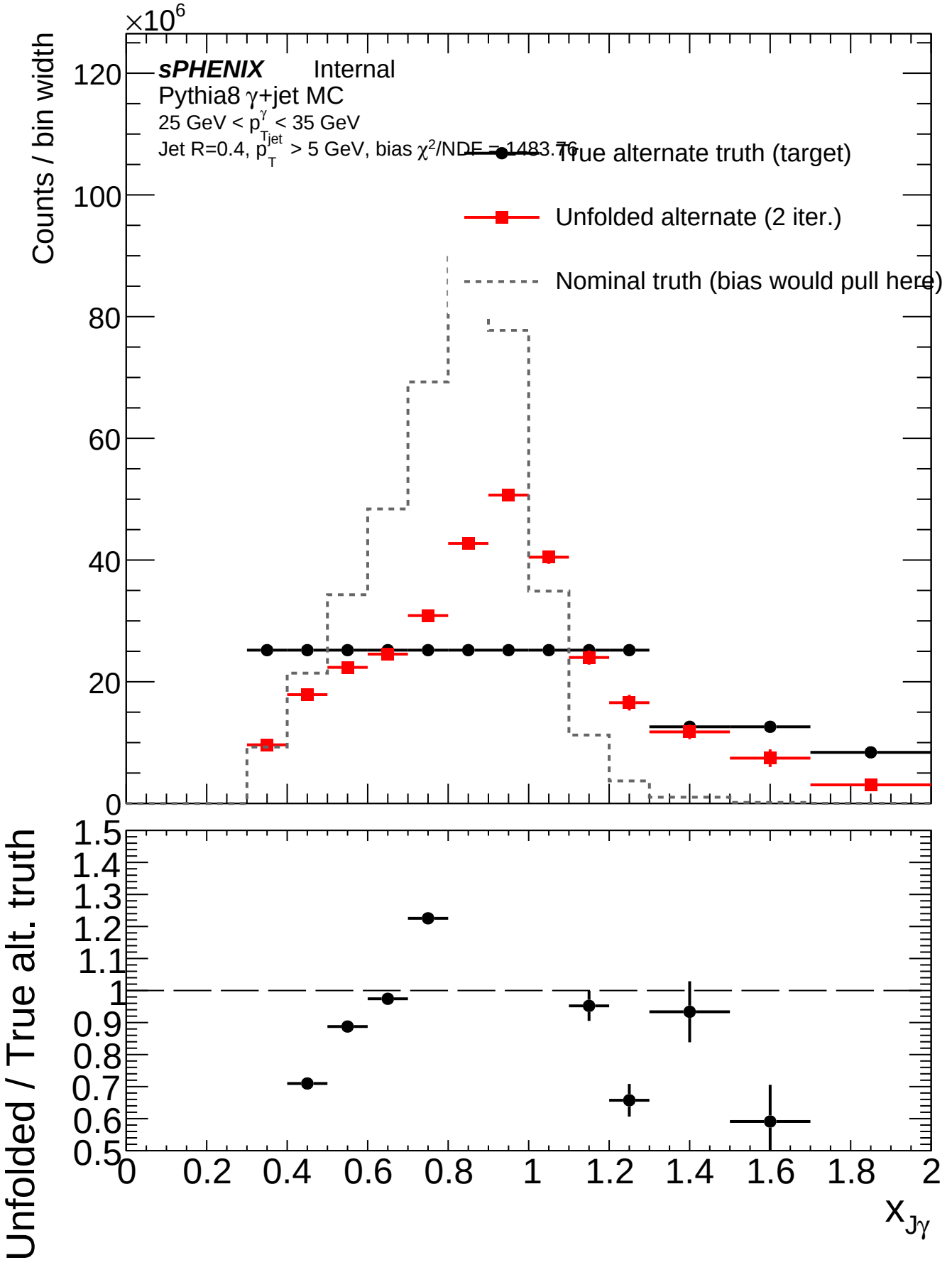

×10^6
Counts / bin width
sPHENIX Internal
Pythia8 γ+jet MC
15 GeV < p_T^γ < 20 GeV
Jet R=0.4, p_T^Tjet > 5 GeV
Both unfolded through the SAME (nominal) response
Unfolded nominal (2 iter.)
Unfolded alternate (2 iter.)
Alternate / Nominal
x_Jγ

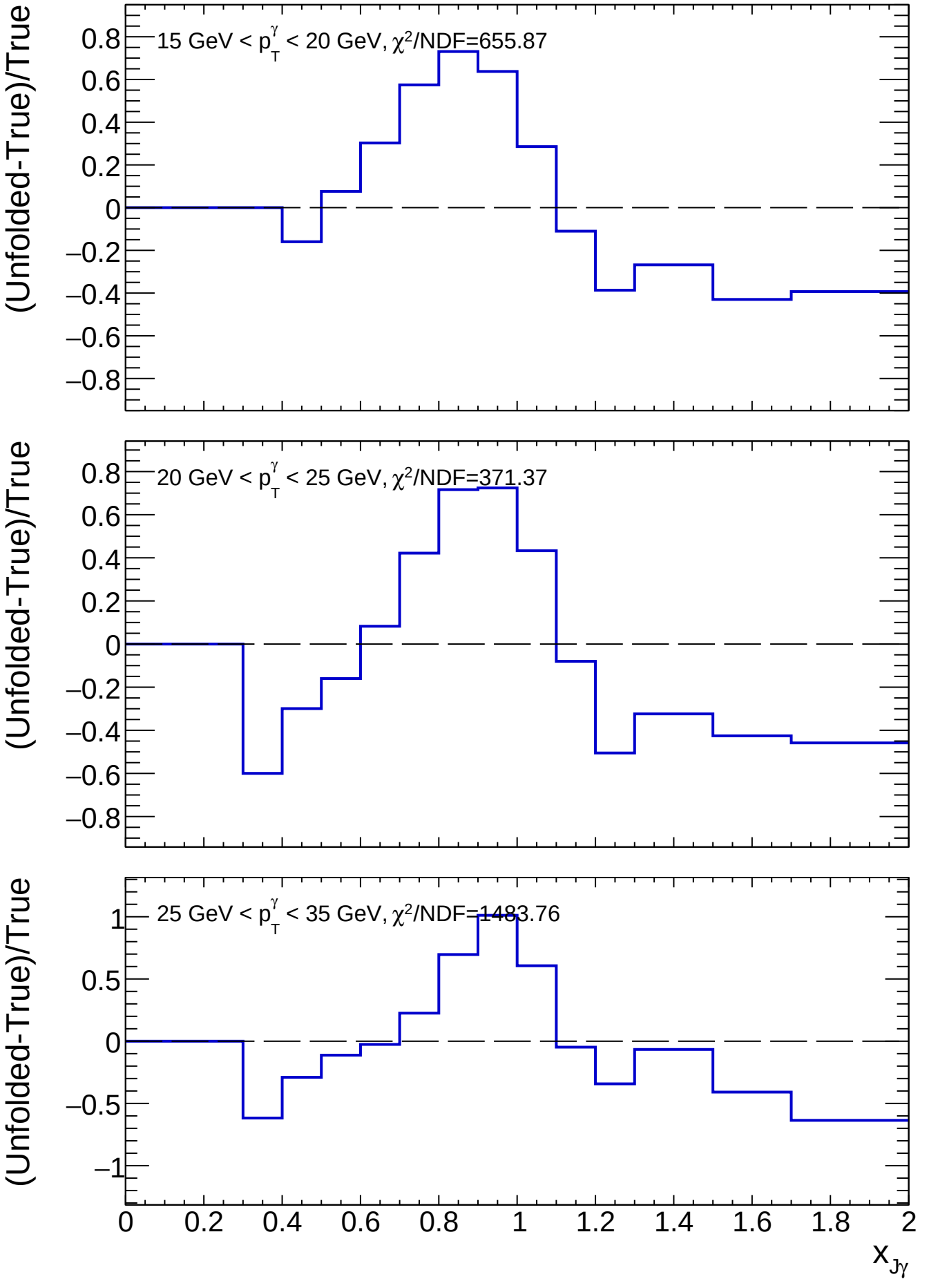

×10^6
sPHENIX Internal
Pythia8 γ+jet MC
20 GeV < p_T^γ < 25 GeV
Jet R=0.4, p_T^Tjet > 5 GeV
Both unfolded through the SAME (nominal) response
Unfolded nominal (2 iter.)
Unfolded alternate (2 iter.)
Counts / bin width
Alternate / Nominal
x_Jγ


×10^6
sPHENIX Internal
Pythia8 γ+jet MC
25 GeV < p_T^γ < 35 GeV
Jet R=0.4, p_T^jet > 5 GeV
Both unfolded through the SAME (nominal) response
Unfolded nominal (2 iter.)
Unfolded alternate (2 iter.)
Counts / bin width
Alternate / Nominal
x_Jγ

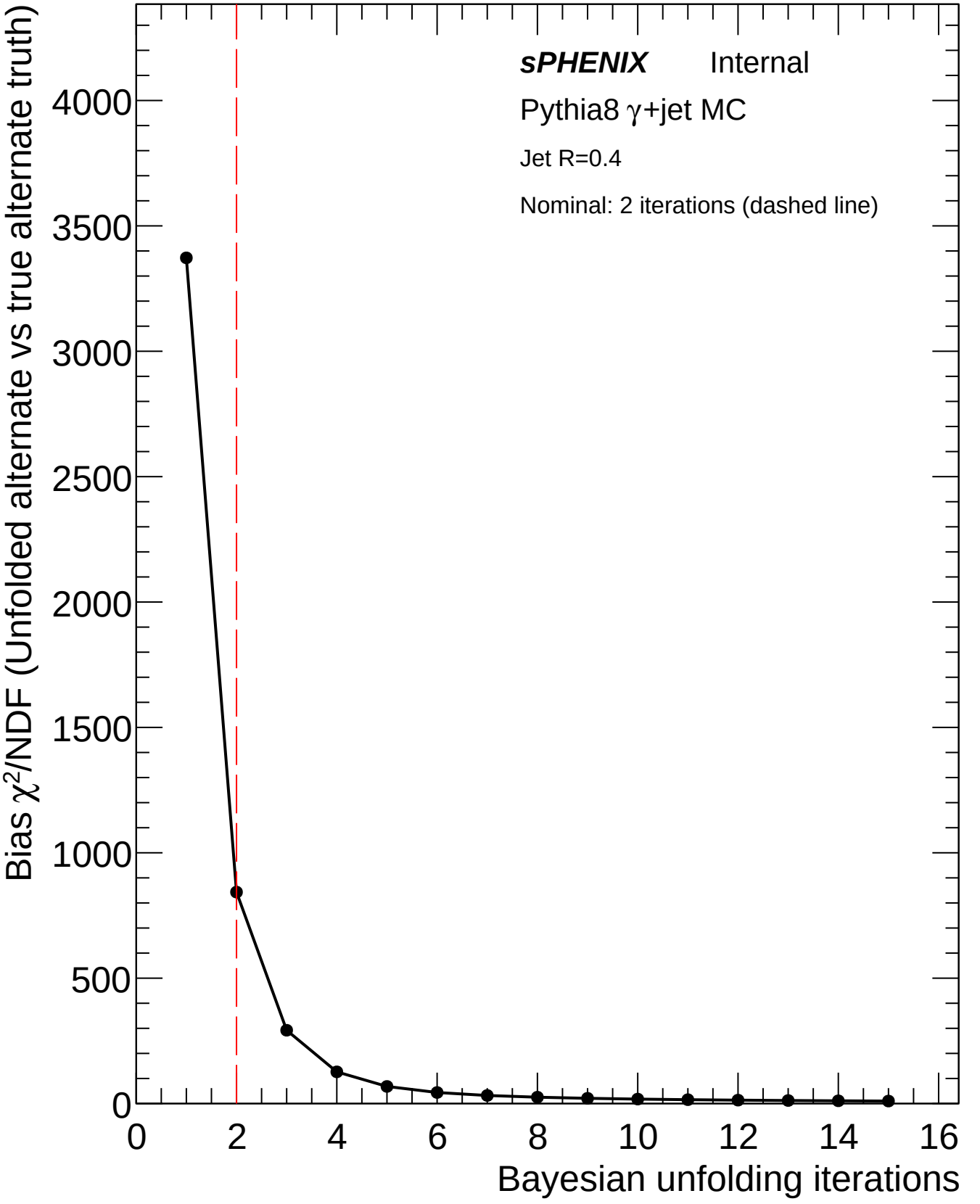

×10^6
sPHENIX Internal
Pythia8 γ+jet MC
15 GeV < p_T^γ < 20 GeV
Jet R=0.4, p_T^Tjet > 5 GeV, bias χ²/NDF = 655.87
True alternate truth (target)
Unfolded alternate (2 iter.)
Nominal truth (bias would pull here)
Counts / bin width
Unfolded / True alt. truth
x_Jγ


×10^6
sPHENIX Internal
Pythia8 γ+jet MC
20 GeV < p_T^γ < 25 GeV
Jet R=0.4, p_T^Tjet > 5 GeV, bias χ²/NDF = 371.37
True alternate truth (target)
Unfolded alternate (2 iter.)
Nominal truth (bias would pull here)
Counts / bin width
Unfolded / True alt. truth
x_Jγ


×10^6
sPHENIX Internal
Pythia8 γ+jet MC
25 GeV < p_T^γ < 35 GeV
Jet R=0.4, p_T^Tjet > 5 GeV, bias
χ²/NDF = 1483.76
True alternate truth (target)
Unfolded alternate (2 iter.)
Nominal truth (bias would pull here)
Counts / bin width
Unfolded / True alt. truth
x_Jγ


15 GeV < p_T^γ < 20 GeV, χ^2/NDF=655.87
20 GeV < p_T^γ < 25 GeV, χ^2/NDF=371.37
25 GeV < p_T^γ < 35 GeV, χ^2/NDF=1483.76
(Unfolded-True)/True
$x_{J\gamma}$

Graph


sPHENIX Internal
Pythia8 γ+jet MC
Jet R=0.4
Nominal: 2 iterations (dashed line)
Bias χ²/NDF (Unfolded alternate vs true alternate truth)
Bayesian unfolding iterations
0
500
1000
1500
2000
2500
3000
3500
4000
0
2
4
6
8
10
12
14
16